



Contributions of soil organic carbon-induced root- and soil properties complexity to water flow in eastern China

Zhiying Tang, Wenqi Zhang, Jinhong Chen, Lu Wang, Yinghu Zhang^{*}

School of Forestry, Nanjing Forestry University, 210037 Nanjing, PR China

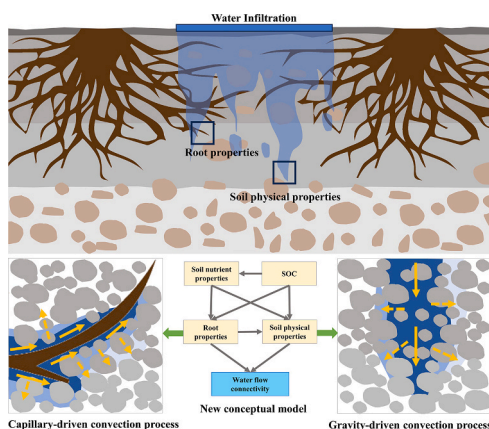
Co-Innovation Center for Sustainable Forestry in Southern China, Nanjing Forestry University, Nanjing 210037, PR China

Jiangsu Provincial Key Laboratory of Soil Erosion and Ecological Restoration, Nanjing Forestry University, Nanjing 210037, PR China

HIGHLIGHTS

- The model of SOC induced root and soil complexity to water flow was constructed.
- Water flow connectivity in each soil depth in different water flow patterns varied.
- SOC by driving roots remarkably dominate convection process in stream buffer zones.
- SOC by driving soil physical property would dominate preferential flow.

GRAPHICAL ABSTRACT



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ABSTRACT

Water flow within the soils affects the efficiency of materials transfer mediated by water. Soil organic carbon (SOC) as an important role in active water flow events can drive the complexity of root-soil synthesis by improving root and soil properties. However, contributions of SOC-induced root- and soil properties complexity to water flow are not well understood. In this study, dye tracing experiments at the three forest stands (oak, pine, and bamboo forests) were conducted to explore water flow patterns, i.e., preferential flow paths (PFP), stream buffer zones (SBZ), and water flow zones (WFZ). X-ray microtomography (CT) scanning was performed to reconstruct the root architecture. The partial least squares path model was applied to quantitatively explore the effects of root- and soil properties on water flow. The results showed that the index of water flow connectivity (*IWFC*) in the PFP and WFZ patterns decreased with increasing soil depth, while *IWFC* in the SBZ pattern increased at first and then decreased. In the PFP pattern, soil physical properties had the larger total effects ($TE = 0.624$) on *IWFC* change compared with root properties ($TE = 0.257$). In the SBZ pattern, the total effects of root properties controlling *IWFC* change ($TE = 0.510$) were greater than soil physical properties ($TE = -0.386$). Both of them can equally affect the *IWFC* in the WFZ pattern. In conclusion, the influences of SOC by driving the

^{*} Corresponding author at: School of Forestry, Nanjing Forestry University, 210037 Nanjing, PR China.

E-mail address: zhang9c415@njfu.edu.cn (Y. Zhang).

changes of soil properties on gravity-driven convective flow process were dramatically stronger than root properties, while SOC could primarily drive the changes of root properties and thereby affect capillary-driven convective flow process. The present results can provide a scientific basis for sustainable forestry management and also a better understanding of the forestry hydrology.

1. Introduction

Forest ecosystem structure can improve the forestry hydrological functions, such as water production, regulation of water balance, and water-related carbon storage (Keles, 2019). The forestry hydrological functions are determined by water availability (Costa et al., 2023). Water availability affects forest ecosystems structure and function by the interaction between water input (i.e., rainfall), water loss (i.e., evapotranspiration and drainage), and water retention, in particular groundwater storage (Costa et al., 2023). For example, forest ecosystems with deep water table depth relying on rainfall inputs, in which root systems are decoupled from the water table, are highly expected to be more hydrologically exposed to drought (Fan et al., 2017). The relationship between forests and water has been studied since the last century (Nainar et al., 2021), and until now, the question upon how to accurately distinguish water flow within the soils has always been a hot topic (Li et al., 2017; Zhang et al., 2023). Water flow within the soils can carry solutes to the different subsoil locations, which results in the spatial heterogeneous water flow patterns (Fuhmann et al., 2019; Zhang et al., 2021). In general, the main patterns of water flow include soil matrix flow and preferential flow (Uchida et al., 2005; Wang et al., 2022). Soil matrix flow is the water flow within the soil matrix, driven solely by capillary forces (Germann and Beven, 1985; Zhang et al., 2017; Jiang et al., 2023). Preferential flow is the water flow along specific preferential paths (i.e., soil macropores, root channels, and soil cracks) bypassing most of the soil matrix. The two patterns of water flow can dramatically affect the hydrological processes (Villamizar and Brown, 2017; Wang et al., 2022).

Water flow in the root-soil synthesis is a basically hydrological process (Beven and Germann, 2013). In the root-soil synthesis, both root systems and soil properties can affect water flow (Jiang et al., 2019; Chen et al., 2021; Zhang et al., 2021), but their relative contribution and importance to water flow are not equal (Jiang et al., 2018; Zhang et al., 2022c; Wang et al., 2023). Soil organic carbon (SOC) may be a key driver of the complex structure of root-soil synthesis. This is due to the special characteristics of SOC and its influence on soil functions. SOC can directly affect soil physical, chemical, and biological properties (Li et al., 2021; Lal, 2021; Védère et al., 2022). In particular, SOC can also directly influence root growth (i.e., root morphology and root biomass) (Lee et al., 2009; Jiang et al., 2021; Georgiou et al., 2022). Therefore, it is dramatically attractive to imply the significant role of SOC in the root-soil interface and its influence on hydrological processes. However, the studies of hydrological responses in root-soil interface are controversial (Li et al., 2023). On the one hand, root systems wedging into the soils can form channels to promote active water flow and infiltration capacity (Zhou et al., 2023). On the other hand, root systems can also reduce permeability, especially in the earliest stages of root growth (Singh et al., 2018). Therefore, the mechanism about SOC-induced root and soil properties on water flow is extremely unknown, and should be got widespread recognition urgently.

Most studies have focused on water flow connectivity on the soil surface, which is defined as “hydrological connectivity” (Bracken et al., 2013; Zhang et al., 2021; Crompton et al., 2023; Wu et al., 2023). The term “hydrological connectivity” defines the water mediated transfer of solutes from one part of the landscape to another at larger scales (Bracken and Croke, 2007). To assess the efficiency and the convenience of water flow within the soils, traditional studies have been conducted by field soil moisture and laboratory water infiltration capacity measurement (Zhu et al., 2023). Maybe water flow connectivity within the

soils will become a promising indicator that defines the convenience level of water flow from one part of the soil layers to another at the soil profile scales (Zhang et al., 2021). Water flow connectivity within the soils is greatly involved in the redistribution of nutrients and energy, influencing the remodeling of soil structure (Cui et al., 2022). It has been gradually recognized in recent years based on its influences on solute transportation, vegetation productivity, and surface water-groundwater recharge (Zhang et al., 2022c). For example, Zhang et al. (2021) assessed the level of water flow connectivity by dye tracing experiments in the Yellow River Delta wetlands and concluded that soil properties showed a higher impact on the connectivity than root systems in the root-soil interface. Zhang et al. (2022a) analyzed the effects of root systems on water flow connectivity in forest soils and found that root length was the most important factor controlling water flow connectivity. However, studies upon how to accurately characterize water flow connectivity within the soils were extremely rare.

Dye tracing experiments are widely used to characterize water flow connectivity within the soils based on dye stained areas. However, these areas might overestimate the areas of actual preferential flow paths. This is because the binary classification method used in previous studies to distinguish preferential flow paths underestimates large and overestimates small dye solution concentrations (Jiang et al., 2019; Zhu et al., 2023). In fact, this method cannot quantify the spatial patterns of dye solution concentrations and tends to sharpen the concentration gradients. In further, it leads to the overestimation of the actual preferential flow paths (Forrer et al., 2000). In this study, dye stained patterns in the soil profiles can be divided into two parts (i.e., preferential flow paths (PFP) and stream buffer zones (SBZ)) by the supervised classification method to avoid the overestimation of actual preferential flow paths (Zhu et al., 2023). The whole dye stained area can be defined as water flow zone (WFZ), which refers to convective flow processes including gravity-driven and capillary-driven convection process. Gravity-driven convection transport pathways, that is PFP, can be described in Kinematic waves approach (Legout et al., 2009; Faeh et al., 2010). Capillary-driven convection process refers to slow water movement through the soil matrix by slow convective flux. The soil matrix in which most of the water are transferred slowly can be treated as SBZ, which can be described by the one-dimensional Richard's equation (Faeh et al., 2010; Jarvis, 2020).

In view of the present study, we conducted the dye tracing experiments at the three forest stands to assess the differences of water flow connectivity within the soils. We hypothesized that SOC-induced root and soil properties complexity contributes differently to the water flow connectivity in the three patterns (PFP, SBZ, and WFZ). The specific objectives of this study were to characterize water flow connectivity in the three water flow patterns and analyze the effects of SOC-induced root and soil properties. Assessing the effects of root and soil properties on water flow connectivity can provide a scientific basis for sustainable forestry management, and can also better understand the forestry hydrology within the soils.

2. Materials and methods

2.1. Study area

The study area is in Jurong City, Jiangsu Province, southeastern China (32°07'N, 119°13'E). It belongs to the north subtropical monsoon climate zone. The forest area is 314.4 ha, the historical average temperature is 15.2 °C, the average annual precipitation is 1048.3 mm, the

average annual relative humidity is 79 % (Li et al., 2020; Zhang et al., 2022a). The soil classification in the study area is silty loam based on the International System of Soil Texture Classification Triangle Coordinate Diagram. The climate is characterized by four distinct seasons, simultaneous rain and heat, and abundant water resources (Zhang et al., 2022a). There are farmlands and plantation forests spread around the study site. Farmers apply a lot of pesticides and fertilizers every year to increase the yield. At the same time, pesticide pest control and chemical drugs addition are needed for the survival of plantation forests (Keleş, 2019). Those foreign additives can affect natural forests through soil water flow (Zhang et al., 2021; Cui et al., 2022). What's more, climate change and increased precipitation in recent decades have also adversely affected forest hydrological functions, ultimately leading to forest ecosystem degradation (Kirilenko and Sedjo, 2007). Therefore, the study of water flow characteristics in soil can better provide us with assistance in determining the degree of threat to forests. We selected three typical subtropical forest stands (oak, pine, and bamboo forests) in the study area, which are all natural secondary forests formed under enclosure protection (Fig. 1).

2.2. Soil physicochemical properties measurement

The experiments were conducted in October 2021. For each forest stand, three sites were set up. Three replicates 20 m × 20 m sampling plots were installed each site. Five-point sampling method was used each plot (Fig. 1B). Three replicates soil cores (height: 10 cm, weight: 7.2 cm, volume: 400 cm³) were excavated at the soil depths of 0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, and 40–50 cm each point. Soil samples inside the soil cores were used to measure soil physical properties (Fig. 1Dc). Soil samples with three replicates were taken from each soil depth to determine soil chemical properties. A total number of 2025 soil samples were collected (3 forest stands × 3 sites × 3 sampling plots × 5 points × 3 replicates × 5 depths). The methods used to determine soil physicochemical properties were shown in Table 1. In this study, soil physical properties included initial soil water content (ISWC), soil porosity, soil bulk density (SBD), and soil texture. Soil texture refers to the content of clay, silt, and sand, and soil porosity includes total soil porosity (TSP), soil capillary porosity (SCP), and soil non-capillary porosity (SNCP). Soil chemical properties include soil organic carbon

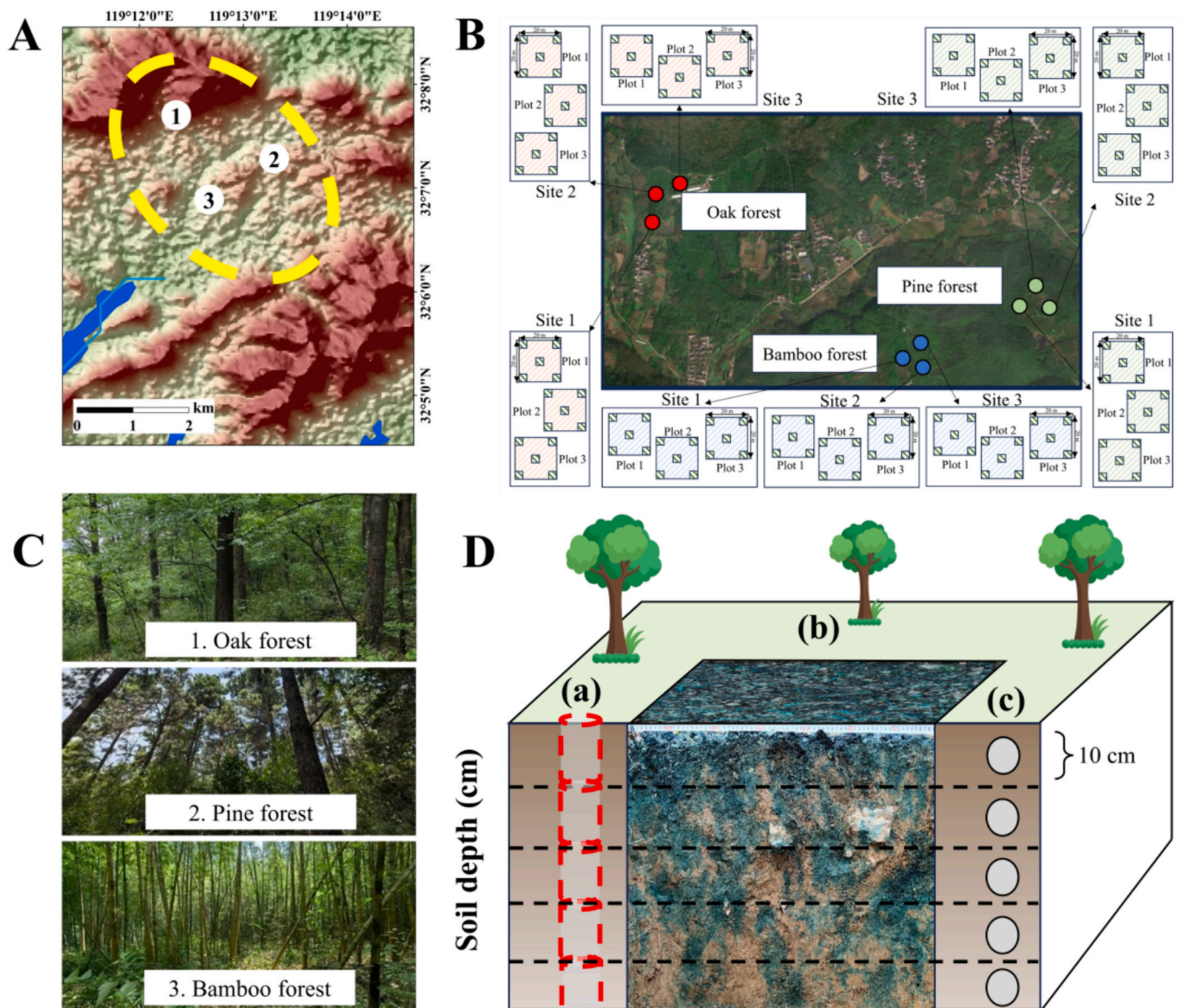


Fig. 1. The study site located in Zhenjiang City, Jiangsu Province, China. A: Locations of (1) oak forest, (2) pine forest, and (3) bamboo forest. B: Three sites were selected in each forest stand, three sample plots were set up in each site, and soil samples were collected by the five-point sampling method. C: Samples of the three forest stands. D: (a) PVC pipe stratified collection of the original soil cores; (b) Dye tracer experiment; (c) Collection of ring cuttings.

Table 1
Methods of determining physicochemical properties of soil.

Soil indicators	Measurement methods	Reference author
Initial soil water content (ISWC), soil bulk density (SBD), total soil porosity (TSP), soil capillary porosity (SCP), soil noncapillary porosity (SNCP)	Ring cutter method	Jiang et al. (2019)
Silt, sand, clay	Aerometer methods (dry sieving and sedimentation analysis)	Zhang et al. (2022a)
Soil organic carbon (SOC)	Dichromate oxidation-colorimetric determination	Sims and Haby (1971)
Soil total nitrogen (TN)	Kjeldahl digestion distillation method	Bremner (1960)
Soil available nitrogen (AN)	Persulfate digestion technique	Sollins et al. (1999)
Soil total phosphorus (TP); soil Available phosphorus (AP)	Continuous flow analytical system (SKALAR SAN ⁺⁺ , The Netherlands)	Miranda et al. (2001)
Soil total potassium (TK); soil available potassium (AK)	Inductively coupled plasma-atomic emission spectrometry (ICPS-7500, Shimadzu, Japan)	Lu (1999)

Note: Soil particle size classification is based on the international soil texture classification standard: clay (<0.002 mm, %), silt (0.02–0.002 mm, %), and sand (2–0.02 mm, %).

(SOC), soil total nitrogen (TN), soil total phosphorus (TP), soil total potassium (TK), soil available nitrogen (AN), soil available phosphorus (AP), and soil available potassium (AK).

2.3. Root properties characterization

To accurately characterize the root properties, a total number of 135 soil cores (height: 10 cm, weight: 7.2 cm, volume: 400 cm³) were collected using PVC pipes with three replicates each soil depth (0–10 cm, 10–20 cm, 20–30 cm, 30–40 cm, and 40–50 cm). We used industrial X-ray CT scanning to image and analyze the root properties (Fig. 1Da). The scanning parameters were set as follows: voxel resolution 61.48 μm, voltage 180 kV, current 160 μA, inter-projection scanning spacing 0.1 mm, exposure time 500 ms, number of stacked sheets 2. The scanned images were manipulated as follows.

(1) The scanned soil cores slices were imported into VGStudio MAX (software VGStudio MAX 2.2, Volume Graphics, Germany) and all slices were converted from 16-bit to 8-bit (Zhang et al., 2022b); (2) the region of interest (ROI) was cropped using the cylindrical cropping tool to prevent the void at the junction of the soil materials and the PVC pipes (Hu et al., 2015); (3) the normalization and 3D median filter plug-ins in the ImageJ software were used to recalculate the gray levels of the images and reduce the noise in the gray images (Liu et al., 2021); (4) the wand tool in Avizo 9.0 software was used to select the root system in the 2D image, continuously adjusting the threshold in the interactive threshold settings to better match the selected root system with the measured root system (Hu et al., 2020); (5) the root system was skeletonized and constructed. The centerline of the root system was set as the skeleton, and the root system was reconstructed to obtain a 3D visualized root network (Luo et al., 2010; Feng et al., 2020). The spatial structural parameters of the root system were calculated using Avizo 9.0 software. The 3D parameters of the root system were quantitatively analyzed based on the number, length, volume, surface area, width, and tortuosity (Zhang et al., 2022b) (Fig. 2). We obtained the root number (RN) by tracing the root branches in Avizo 9.0 software (Perret et al., 2007), and those parameters were calculated as follows.

$$\bullet \text{Root length (RL, cm)} = \sum_{i=1}^n l_i \quad (1)$$

where l_i is equal to the Euclidean between two points Reid's distance

(μm).

$$\bullet \text{Root volume (RV, cm}^3\text{)} = N \times V_0 \quad (2)$$

where V_0 is the minimum volume of volume element units and N is the number of volume elements contained in the three-dimensional root system.

$$\bullet \text{Root surface area (RSA, cm}^2\text{)} = N \times A_0 \quad (3)$$

where A_0 is the minimum pixel cell area and N is the total number of pixels in the area covered by the root system.

$$\bullet \text{Root tortuosity (RT)} = \frac{L_T}{L_0} \quad (4)$$

where L_T is the total length of the actual path of the pore throat and L_0 is the directrix distance between the two ends of the pore throat.

We calculated the root width (RW), which is equal to the diameter of the average internal tangent bulb of the root.

Then, we divided RL, RV, RSA, and RW by RN to obtain root length density (RLD, per unit cm), root volume density (RVD, per unit cm³), root surface area density (RSAD, per unit cm²), and root mean width (RMW, per unit cm).

2.4. Water flow connectivity quantification

The dye tracing method was used to obtain the spatial distribution of water flow paths within the soils (Flury et al., 1994; Jiang et al., 2021; Zhang et al., 2021). Nine experimental plots each forest stand were selected independently (Wang et al., 2022). The procedure was as follows: (1) a hollow square iron frame (0.7 × 0.7 × 0.7 m) was inserted into the soil at a depth of about 0.2 m. The soil on both sides of the iron frame was compacted tightly to avoid boundary effects of the dye solution; (2) a rainfall simulator was used to conduct precipitation events, and 50 L (4 g L⁻¹) Brilliant Blue FCF (C.I. Food Blue 2) was sprayed uniformly into the iron frame; (3) the iron frame was covered with plastic film for 24 h after application of the dye solution to avoid interference from evaporation and precipitation; (4) vertical soil profiles with a maximum dyeing depth of 0.5 m were excavated and then photographed with a high-precision digital camera (Fig. 1Db).

After obtaining dye stained patterns of the vertical soil profiles, geometric corrections and dye stained areas extraction were made using Adobe Photoshop CC 2018 software to get dye stained images. Then, the dye stained images were imported into ENVI 5.3 software for supervised classification (Zhu et al., 2019; Jiang et al., 2019; Zhang et al., 2023). The dye stained areas in this study were categorized into two parts: heavy dye stained areas and low dye stained areas. The heavy dye stained areas indicate the more active areas of water flow (preferential flow paths) with a high dye stained concentration of >2.0 g L⁻¹, while the low dye stained areas indicate the areas where the water flow interacts with the surrounding soil matrix (stream buffer zones) with low dye stained concentration < 2.0 g L⁻¹. In addition, the WFZ refers to convective flow zones including both PFP and SBZ. That is, the combination of the heavy- and the low dye stained areas in the soil profiles are defined as WFZ (Fig. 3) (Zhu et al., 2019; Jiang et al., 2021; Chen et al., 2021). We used the sharpening tool of ENVI 5.3 software to reduce the noise in the images (Jiang et al., 2019; Zhang et al., 2023). The binarization of the supervised classified images was carried out by the ImageJ 1.8.0 Fiji software, and the dye coverage (DC) and fractal dimension (FD) of the images were counted (Zhang et al., 2021). The FD is an indicator applied to characterize complexity and scale-independent properties of an object (Guo et al., 2020). In this study, a higher FD at the same DC means a more complex dye-stained patches network and also a structural pore complexity (Guo et al., 2020). Although DC of the soil profiles is same, the pore connectivity which can reflect how much time to balance the hydraulic pressure between the dye-stained areas

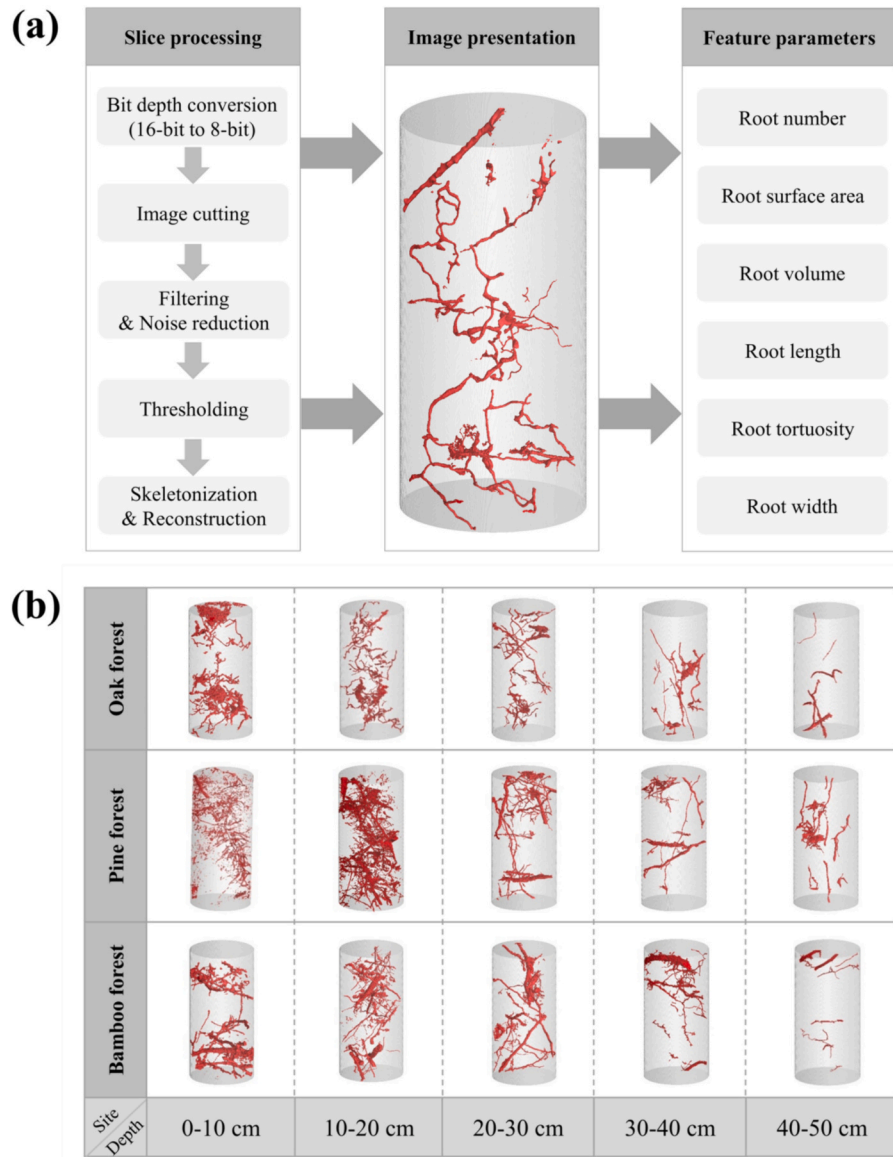


Fig. 2. CT scanning of the soil cores to reconstruct the root systems, the parameters obtained, and the distribution status of the root systems in the soil. a: Three-dimension (3D) reconstruction of the root to obtain root feature parameters. b: Samples of the root 3D configuration of the three forest stands.

and non-dye-stained areas may not be equal due to the fragmentation of dye-stained patches (Zhang et al., 2021, 2022c). This is because pore connectivity can reflect the degree of the fractal dimension in terms of pore distribution. For example, a higher FD of pore distribution at the same soil porosity implies a more complex distribution of pore network given the greater pore-wall surface area, which delays gravity-driven convective process because the large capillary-driven convective process increases the residence time of water (Guo et al., 2020). Thus, the increase or decrease of FD in terms of pore distribution provides an indication of increased or decreased resistance to flow. The formulas are as follows:

$$\bullet \text{DC (\%)} = [D / (D + ND)] \cdot 100\% \quad (5)$$

where D is the dye-stained area, and ND is the non-dye-stained area.

$$\bullet \text{FD} = \lim_{\varepsilon \rightarrow 0} [\log N(\varepsilon) / \log 1/\varepsilon] \quad (6)$$

where ε is the average width of each dye-stained area, and $N(\varepsilon)$ is the number of fractals covered by this dye-stained area.

We coupled DC and FD to obtain the index of water flow connectivity

(IWFC), which was in the range [0,1] (Zhang et al., 2021).

$$\bullet \text{IWFC} = \frac{\sum_{i=1}^n \frac{DC_i}{FD_i}}{n} \quad (7)$$

where n is the number of dye-stained horizontal profiles at a certain soil depth.

2.5. Representative indicator selection

To remove redundant data, the results of principal component analysis (PCA) and correlation analysis were regarded as a reference to identify the key factors affecting IWFC (Zhang et al., 2021). In this study, the indicators in the total dataset (TDS) include soil and root properties. Soil properties include soil physical properties (ISWC, SBD, TSP, SCP, SNCP, Clay, Silt, and Sand) and soil chemical properties (TN, TP, TK, AN, AP, and AK), while root properties include RN, RW, RL, RSA, RV, RT, RMW, RLD, RSAD, and RVD. We constructed the minimum dataset (MDS) based on the key indicators selected through PCA and correlation analysis. After PCA, principal components (PC) with a characteristic

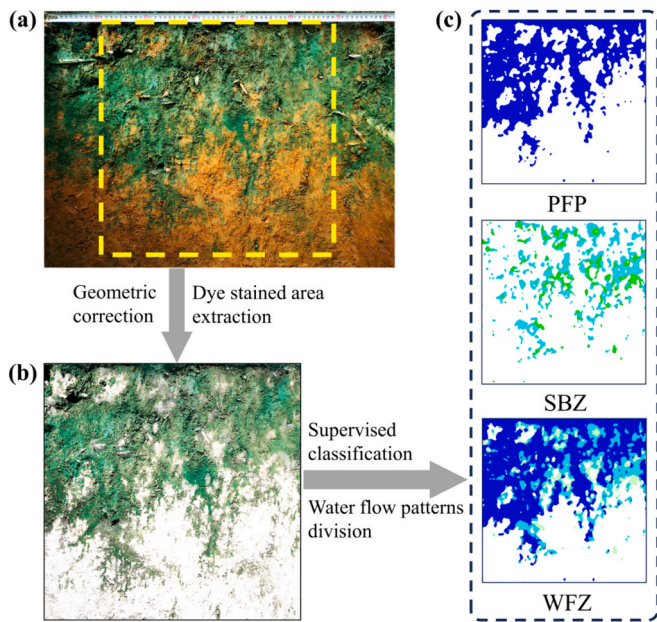


Fig. 3. Processing of dyeing images and delineation of water flow patterns. a: The original image of the dye stained area (the 50 × 50 cm area in the middle yellow boxed part was cropped to prevent edge effects). b: The image after dye stained area extraction. c: The three water flow patterns delineated.

value >1 was selected. In each PC, only indicators with absolute load values within 10 % of the maximum factor load were selected. Next, Pearson correlation analysis was used to determine whether other indicators should be removed when more than one indicator were retained in the PC (Li et al., 2024).

2.6. Data analysis

Normality of the data was determined using Shapiro-Wilktest. One-way ANOVA in SPSS 26.0 (Statistical Graphics Crop, Princeton, USA) was used to compare the differences in *IWFC* between the three forest stands and different soil depths in the same forest type. The results were statistically significant at the 5 % significance level. Pearson correlation was used to analyze the relationships among soil physical properties, SOC, soil chemical properties, root properties, and the water flow

connectivity. Random forest analysis in R 4.3.0 (<https://www.r-project.org/>) can be used to identify the main factors that affect the connectivity. We used the packages: “linkET”, “randomForest”, “corrplot”, “ggplot2”, “vegan”, and “dplyr”. The relationships among soil physical properties, soil chemical properties, and root properties were visualized by principal component analysis (PCA) using the plug-in “Principal Component Analysis” of Origin 2022 (<https://www.originlab.com/>). To determine the direct and indirect effects of SOC-induced root and soil properties on the water flow connectivity (Feng et al., 2023), partial least squares path model (PLS-PM) by Smart PLS 3.2.9 software (<https://www.smartpls.com/>) was developed.

3. Results

The minimum dataset (MDS) based on soil and root properties were constructed in Table 2 and Fig. 4. The results of MDS showed that TSP, ISWC, Silt, TN, AK, RN, and RMW were selected to the representative indicators which mainly affect water flow connectivity.

3.1. *IWFC* in the PFP pattern and its correlation with root and soil properties

From Fig. 5, *IWFC* in the PFP pattern decreased with increasing soil depth at the three forest stands. *IWFC* in the whole soil profiles decreased with the following order: bamboo forests (0.269 ± 0.045) > pine forests (0.227 ± 0.023) > oak forests (0.206 ± 0.035). *IWFC* of the total soil profile in the bamboo forest was significantly different with the oak forest ($P < 0.05$), while not significantly different with the pine forest ($P > 0.05$). *IWFC* in the pine forests was not significantly different with that in the oak forest ($P > 0.05$).

SBD was significantly negative correlated with *IWFC* ($P < 0.001$), while TPS, SCP, SNCP, TN, AN, AP, SOC, and RSA were significantly positive correlated with *IWFC* ($P < 0.05$). The other parameters of root and soil properties showed no significant correlations with *IWFC* ($P > 0.05$) (Fig. 6a). The results from Random Forest Analysis showed that the ranking might be partly explained by a combination of SOC, soil physical properties (SBD, TSP, and SCP), and soil chemical properties (AP, TN, and AN) and that those factors exerted a larger influence on *IWFC* than root properties (Fig. 6c). However, the ranking of SOC, soil physical properties, soil chemical properties, and root properties in the Random Forest Analysis explained the predictive power of those variables on *IWFC*, but didn't quantify their effects. From the analysis of the PLS-PM, soil physical properties and root properties as direct factors

Table 2

Changes in soil physicochemical properties and root system properties with soil depth in three forest stands (oak, pine, and bamboo forests) selected by principal component analysis and correlation analysis.

Study sites	Depth (cm)	SOC (g kg ⁻¹)	Soil physical properties			Soil chemical properties		Root properties	
			ISWC (%)	TSP (%)	Silt (%)	TN (g kg ⁻¹)	AK (mg kg ⁻¹)	RN	RMW (per unit cm)
Oak forests	0–10	16.54 ± 0.05	11.79 ± 1.08	48.15 ± 0.75	88.14 ± 9.11	1.55 ± 0.02	86.82 ± 48.90	7630.00 ± 5425.98	0.16 ± 0.02
	10–20	6.56 ± 0.06	9.56 ± 0.81	42.71 ± 1.53	83.53 ± 10.94	0.74 ± 0.03	62.20 ± 34.61	3476.67 ± 1962.89	0.25 ± 0.09
	20–30	4.12 ± 0.18	9.87 ± 0.46	39.40 ± 1.07	81.35 ± 7.67	0.58 ± 0.03	63.89 ± 37.54	4188.00 ± 4027.78	0.19 ± 0.03
	30–40	2.72 ± 0.36	9.95 ± 0.60	37.78 ± 0.78	87.46 ± 7.73	0.46 ± 0.02	62.16 ± 36.98	2046.33 ± 1978.84	0.15 ± 0.04
	40–50	2.57 ± 0.22	10.53 ± 1.77	37.43 ± 0.74	83.16 ± 4.78	0.39 ± 0.01	59.24 ± 32.14	456.33 ± 147.11	0.17 ± 0.07
Pine forests	0–10	17.13 ± 3.77	23.53 ± 2.90	55.08 ± 1.59	89.43 ± 5.47	1.58 ± 0.07	83.73 ± 41.67	3370.00 ± 731.73	0.16 ± 0.06
	10–20	7.97 ± 0.75	22.68 ± 2.10	50.09 ± 2.21	84.45 ± 3.06	0.92 ± 0.10	61.16 ± 30.12	8114.67 ± 5246.03	0.20 ± 0.12
	20–30	4.24 ± 0.54	22.05 ± 1.69	47.64 ± 3.59	91.43 ± 2.59	0.64 ± 0.08	77.74 ± 43.53	2409.00 ± 1848.20	0.09 ± 0.02
	30–40	3.81 ± 0.38	21.28 ± 1.40	42.40 ± 2.66	81.53 ± 5.40	0.60 ± 0.03	97.50 ± 62.48	1987.67 ± 513.74	0.12 ± 0.03
	40–50	2.95 ± 0.35	20.22 ± 1.30	40.84 ± 2.82	87.42 ± 4.87	0.51 ± 0.06	109.99 ± 78.91	638.00 ± 60.85	0.19 ± 0.03
Bamboo forests	0–10	10.43 ± 1.45	19.37 ± 4.49	53.86 ± 5.77	76.92 ± 5.88	1.16 ± 0.16	75.06 ± 39.15	1881.67 ± 1287.83	0.32 ± 0.04
	10–20	6.61 ± 1.36	19.18 ± 1.60	51.15 ± 4.98	81.95 ± 2.15	0.79 ± 0.07	61.15 ± 37.91	1593.00 ± 449.05	0.26 ± 0.06
	20–30	4.01 ± 1.90	15.60 ± 5.90	49.97 ± 5.36	78.86 ± 7.47	0.55 ± 0.15	71.58 ± 38.68	2018.00 ± 963.09	0.24 ± 0.02
	30–40	3.99 ± 2.67	20.71 ± 6.50	47.84 ± 5.43	85.70 ± 4.45	0.46 ± 0.04	77.81 ± 40.61	1414.00 ± 1128.03	0.18 ± 0.08
	40–50	2.25 ± 0.65	24.30 ± 5.10	41.67 ± 0.97	74.29 ± 10.88	0.45 ± 0.08	96.45 ± 60.35	812.67 ± 729.39	0.18 ± 0.02

SOC: Soil organic carbon; ISWC: Initial soil water content; SBD: Soil bulk density; TSP: Total soil porosity; TN: Soil total nitrogen; AK: Soil available potassium; RN: Root number; RMW: Root mean width.

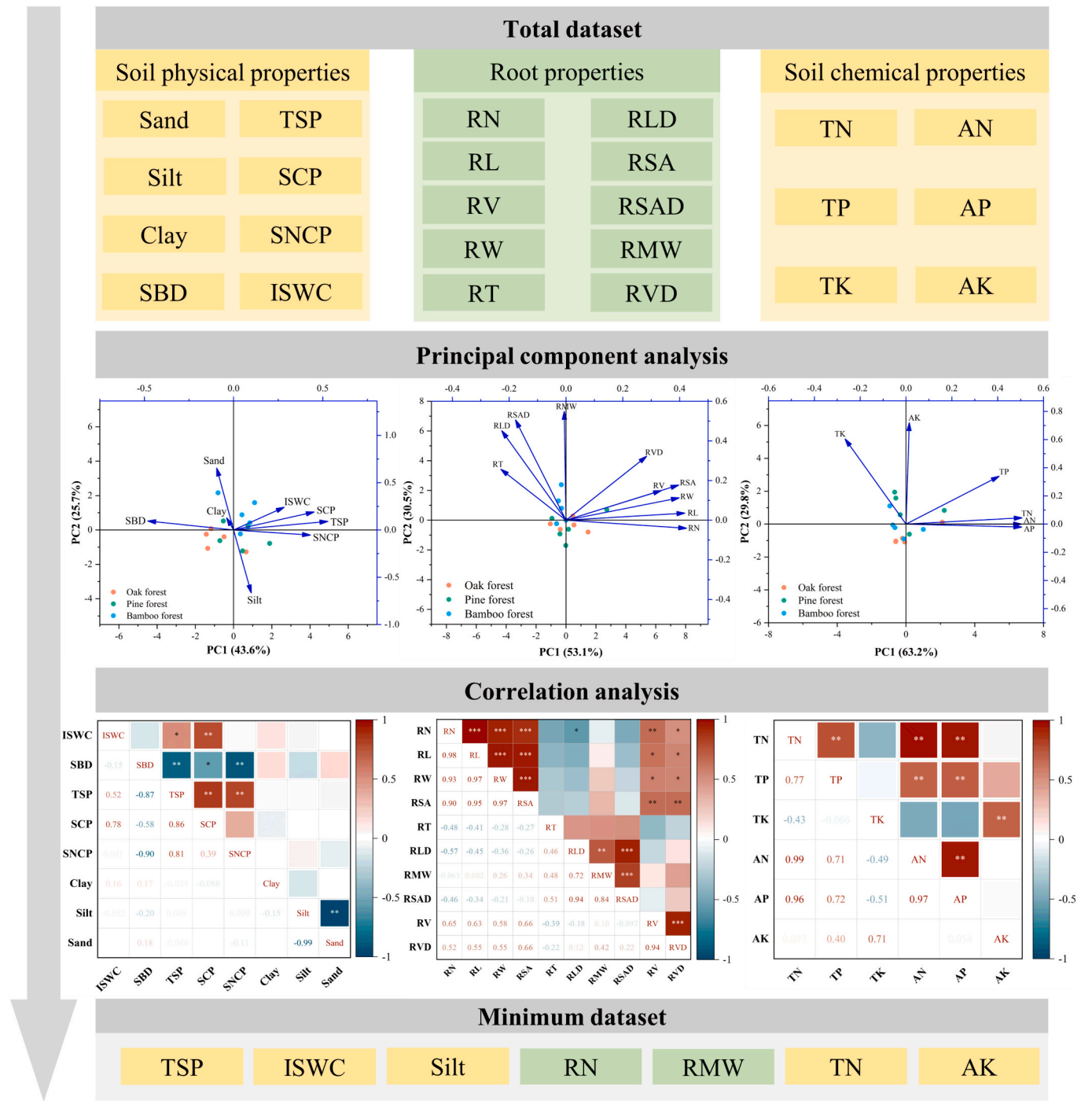


Fig. 4. The process of principal component analysis (PCA) and correlation analysis of the total dataset (TDS) to construct the minimum dataset (MDS). ISWC: Initial soil water content; SBD: Soil bulk density; TSP: Total soil porosity; SCP: Soil capillary porosity; SNCP: Soil non-capillary porosity; TN: Soil total nitrogen; TP: Soil total phosphorus; TK: Soil total potassium; AN: Soil available nitrogen; AP: Soil available phosphorus; AK: Soil available potassium; RN: Root number; RL: Root length; RSA: Root surface area; RW: Root width; RV: Root volume; RT: Root tortuosity; RLD: Root length density; RSAD: Root surface area density; RMW: Root mean width; RVD: Root volume density.

controlling *IWFC* in the PFP pattern showed unequal influences. The analysis of PLS-PM showed that soil physical properties had a higher direct effect ($DE = 0.624$) than root properties ($DE = 0.265$) (Fig. 7a, b). However, SOC and soil chemical properties as indirect factors could affect *IWFC* change. In the PFP pattern, soil physical properties had the largest total effects ($TE = 0.624$) on *IWFC* change, followed by SOC ($TE = 0.484$), soil chemical properties ($TE = 0.344$), and root properties ($TE = 0.257$). The indirect effect of SOC on water flow connectivity through soil properties ($IE = 0.371$) was highly larger than that through root

properties ($IE = 0.123$) (Fig. 7a, b).

3.2. *IWFC* in the SBZ pattern and its correlation with root and soil properties

IWFC in the SBZ pattern increased at first and then decreased with increasing soil depth (Fig. 5). *IWFC* in the whole soil profiles decreased with the following order: oak forests (0.185 ± 0.057) > pine forests (0.119 ± 0.033) > bamboo forests (0.108 ± 0.051). *IWFC* in the oak

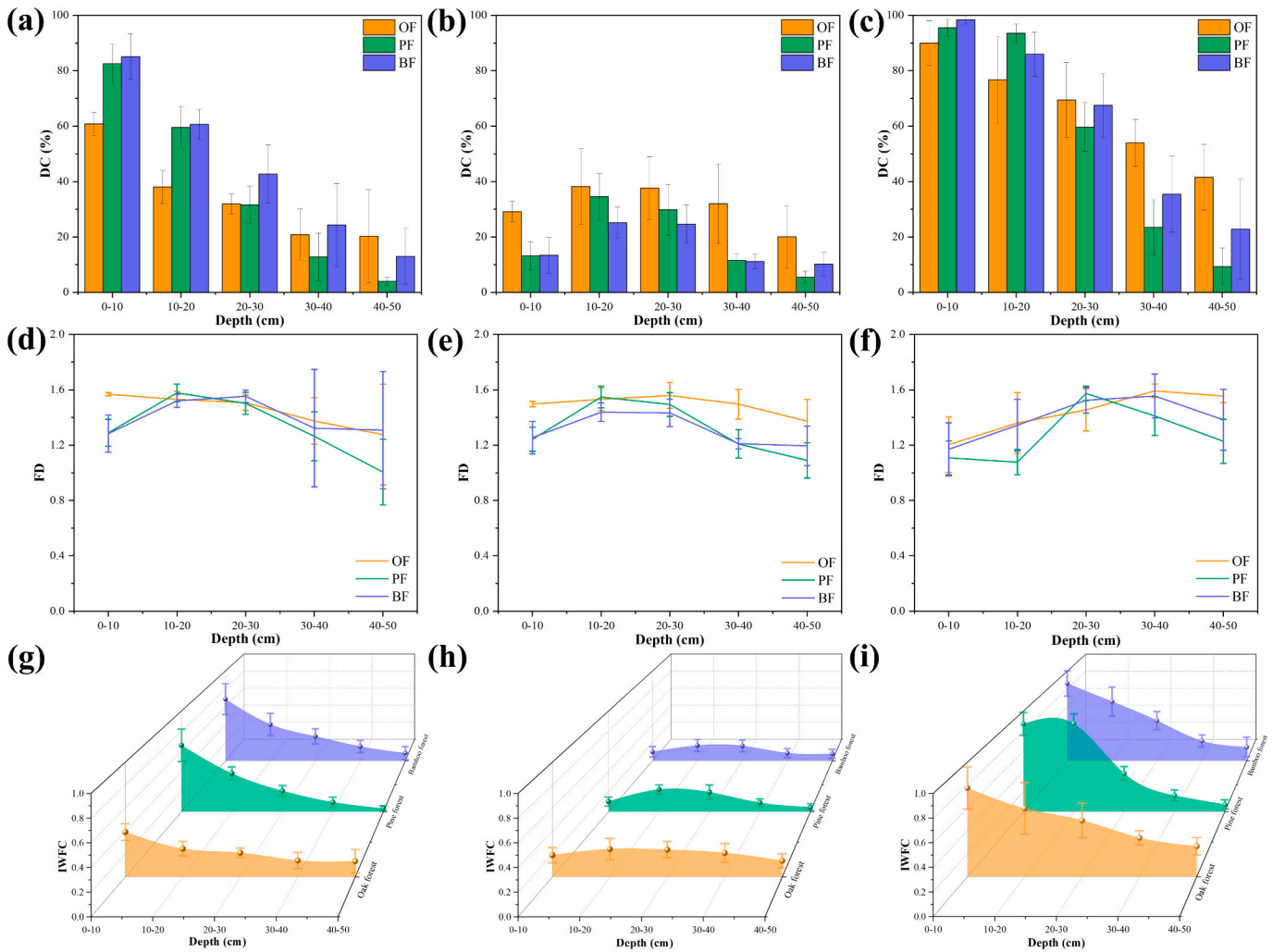


Fig. 5. Variation of dye coverage (DC), fractal dimension (FD), and the index of water flow connectivity (IWFC) with soil depth for three water flow patterns (PFP, SBZ, and WFZ) in three stands (oak, pine, and bamboo forests). a, d, and g represent DC, FD, and IWFC in the PFP pattern; b, e, and h represent DC, FD, and IWFC in the SBZ pattern; d, f, and i represent DC, FD, and IWFC in the WFZ pattern, respectively. OF: oak forest; PF: pine forest; BF: bamboo forest.

forests was significantly different with that in the pine forests and the bamboo forests ($P < 0.05$), while IWFC in the pine forests was not significantly different with that in the bamboo forests ($P > 0.05$).

TK, AK, and ISWC were significantly negative correlated with IWFC ($P < 0.05$), while RN, RL, RW, and RSA were significantly positive correlated with IWFC ($P < 0.05$). The other parameters of root and soil properties showed no significant correlations with IWFC ($P > 0.05$) (Fig. 6a). The ranking of the relative importance of those variables from Random Forest Analysis showed that IWFC might be primarily explained by root properties (RN, RT, RW, RV, RSA, and RL) and that those factors exerted a larger influence on IWFC than soil physical properties (Fig. 6c). From the analysis of the PLS-PM, soil physical properties and root properties as direct factors controlling IWFC in the SBZ pattern showed that root properties had positive direct effects on IWFC, while soil physical properties had negative direct effects. Root properties had a higher direct effect ($DE = 0.444$) than soil physical properties ($DE = -0.386$) (Fig. 7c, d). The total effects of root properties, soil physical properties, soil chemical properties, and SOC controlling IWFC change were 0.510, -0.386 , -0.100 , and 0.023, respectively. SOC's indirect effects on the water flow connectivity through root properties ($IE = 0.249$) was higher than that through soil properties ($IE = -0.036$) (Fig. 7c, d).

3.3. IWFC in the WFZ pattern and its correlation with root and soil properties

Fig. 5 showed that IWFC in the WFZ pattern decreased with increasing soil depth. IWFC in the whole soil profiles decreased with the following order: oak forests (0.408 ± 0.049) > bamboo forests (0.373 ± 0.046) > pine forests (0.340 ± 0.020). IWFC of the whole soil profile in the three forest stands didn't show significantly difference with each other ($P > 0.05$). IWFC of the whole soil profiles in the oak forests was significantly different with that in the pine forests and the bamboo forests ($P < 0.05$), while IWFC in the pine forests was not significantly different with that in the bamboo forests ($P > 0.05$).

SBD and TK were significantly negative correlated with IWFC ($P < 0.01$), while TSP, SCP, SNCP, TN, AN, AP, SOC, RN, RL, RW, RSA, and RVD were significantly positive correlated with IWFC ($P < 0.05$). The other parameters of root and soil properties showed no significant correlations with IWFC ($P > 0.05$) (Fig. 6a). The results from Random Forest Analysis showed that IWFC might be partly explained by a combination of SOC, root properties (RW, RV, RL, and RSA), soil physical properties (SBD and TSP), and soil chemical properties (AP, AN, and TN) (Fig. 6c). The analysis of the PLS-PM showed that soil physical properties and root properties had nearly equal direct effects on IWFC change, which were 0.422 and 0.459, respectively (Fig. 7e, f). SOC had the highest total effect ($TE = 0.483$) on IWFC change compared with soil physical

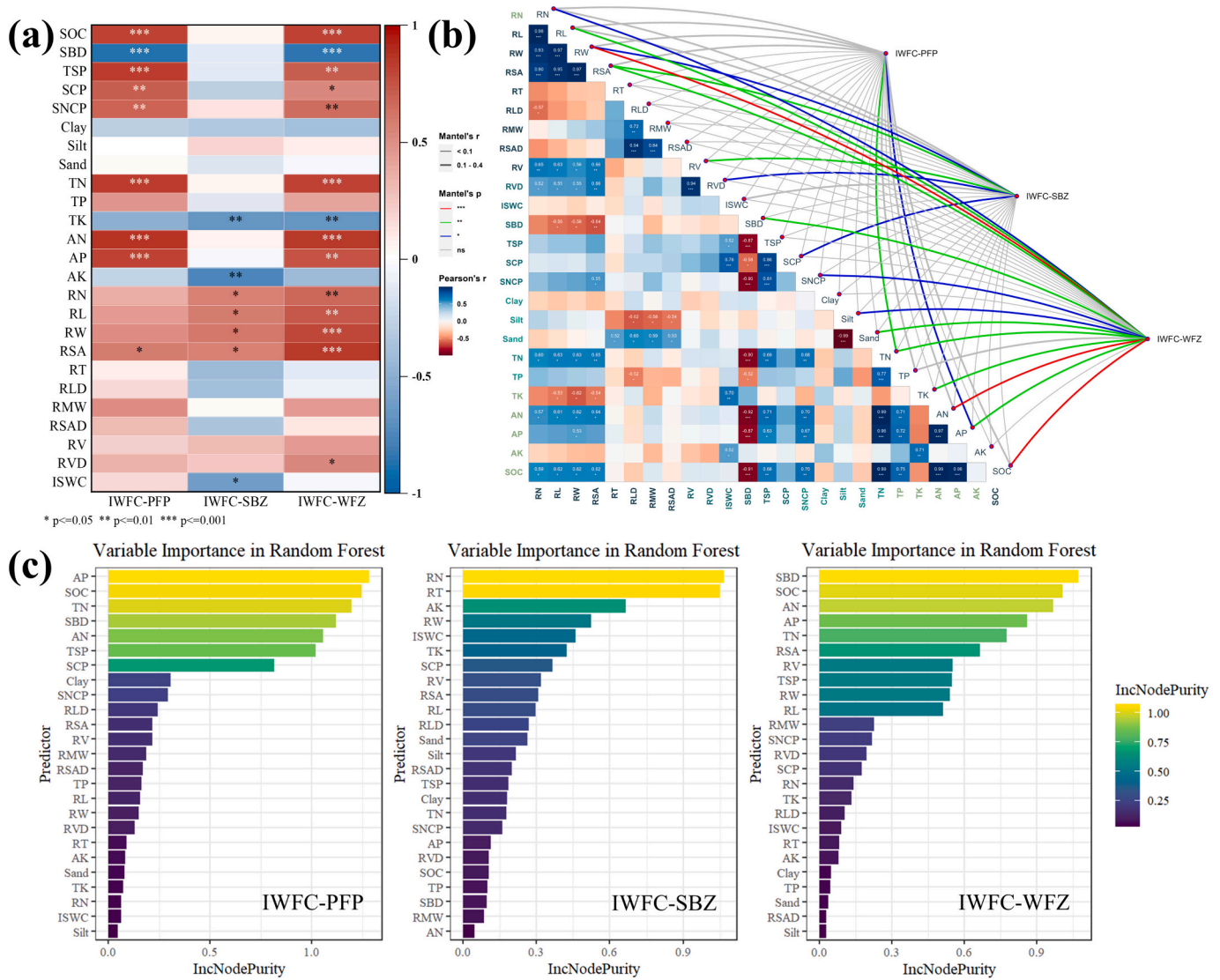


Fig. 6. Correlation, mental-test and random forest analyses of the relationship between soil physicochemical properties and water flow connectivity in the three water flow patterns (PFP, SBZ, and WFZ). a: Correlation analysis between soil physicochemical properties and root properties on *IWFC* in the three water flow patterns. b: Correlation analysis between soil properties and root properties, mantel test of the root and soil properties on *IWFC* in the three water flow patterns. c: Random forest analysis of the importance of soil physicochemical properties and root properties on *IWFC* for the three water flow patterns.

properties (TE = 0.422), root properties (TE = 0.439), and soil chemical properties (TE = 0.312). The indirect effect of SOC on the water flow connectivity through soil properties (IE = 0.344) was larger than that through root properties (IE = 0.217) (Fig. 7e, f).

4. Discussion

4.1. Root properties and water flow

In agreement with the results of Cui et al. (2022), Zhang et al. (2022c), and Cai et al. (2024), RN and RMW were selected as the representative root parameters controlling *IWFC* in the present study (Fig. 4). One possible explanation might be that RN can shape soil pore structure and alleviate soil compaction, which positively affects soil penetration (Benard et al., 2021). Another possible explanation might be that RMW directly affects the degree of the contact between root systems and the surrounding soil matrix. The rough surface of coarse root systems increases the resistance to the dye transportation. In addition, the contact also creates a heterogeneous environment, which causes “water accumulation” during the dye transportation process (Zhang et al.,

2022c).

In the present study, the results showed that root properties positively exerted a higher direct effect on the water flow connectivity in the SBZ pattern compared with that in the PFP pattern (Fig. 7a, b), which is supported by the results of Wang et al. (2022). It might be caused that root growth can create compressive stress on the surrounding soil particles (Pierret et al., 2016; Jin et al., 2017). Due to the creation of the high compressive stress, root systems can deform the soils by displacing soil particles (Jin et al., 2013, 2017). In further, formation of a new pore, elongation along an existing pore, and negotiation through an existing pore can result in the higher complexity of the dense mesh structures of pores (Liu et al., 2014; Hao et al., 2021). Additionally, the complex mesh structures of pores might extremely promote the stream buffer zone effect, which is similar to the results of Jiang et al. (2021). In this study, the largest RN in the oak forest (Table 2) during root growth can generate laterally and longitudinally heterogeneous root systems in the soil profiles, which can form more complex pore networks, and in further result in a higher stream buffer zone effect (Fig. 5). Another possible explanation is that root systems interaction on the surrounding soil particles will increase cohesiveness or enhance the aggregate

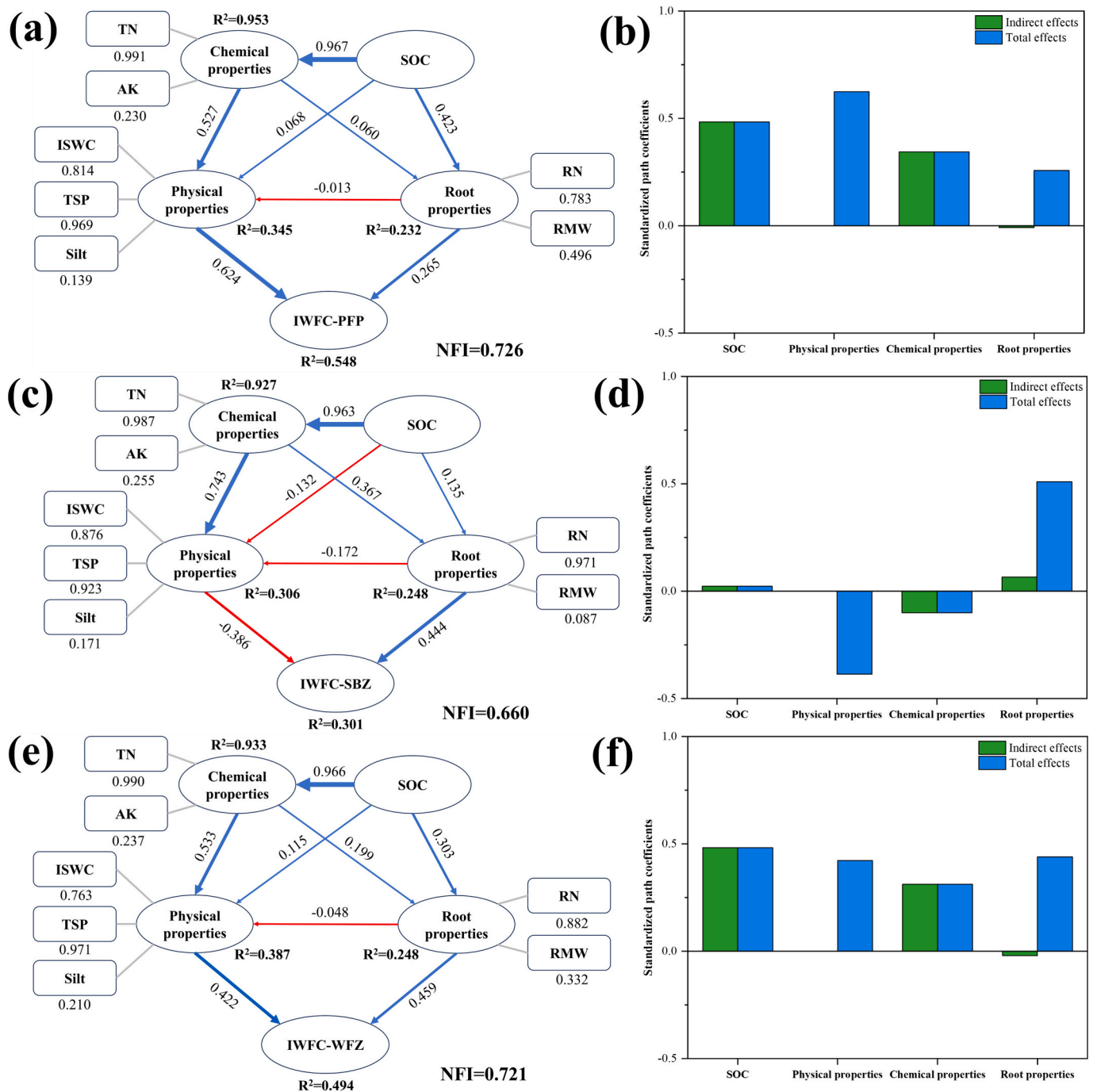


Fig. 7. Partial least squares path model (PLS-PM) shows the direct and indirect effects of the SOC-induced root and soil properties on the water flow connectivity in the three water flow patterns (PFP, SBZ, and WFZ). The width of the arrows is proportional to the path coefficients. Red and blue arrows indicate reverse and forward causality, respectively. Numbers on arrows indicate effective standardized path coefficients. R^2 indicates the variance of the dependent variable explained by the model. NFI indicates normed fit index. a, c, and e show the standardized paths of the SOC-induced root and soil properties for the water flow connectivity in the PFP, SBZ, and WFZ patterns, respectively. b, d, and f show the standardized path coefficients. Soil physical properties include ISWC, TSP, and silt; Soil chemical properties include TN and AK; Root properties include RN and RMW.

stability, resulting in soil fine particles formation (Wang et al., 2021; Wu et al., 2023; Xiao et al., 2023). Root systems can directly reinforce soils by physically binding and connecting soil fine particles (Poirier et al., 2018; Hao et al., 2021), which can promote the lateral infiltration of the dye into the stream buffer zones (Basset et al., 2023). In the present results (Table 2), the higher fraction of soil fine particles reinforced by root systems in the oak/pine forests compared with that in the bamboo forest can support the hypothesis. Consequently, root exudates might create additional capillary forces that retain water and thereby promote

the lateral infiltration, which agrees with the results of Carminati et al. (2017).

In agreement with the results of Zhang et al. (2022c) and Luo et al. (2023), the effects of root systems on active water flow, i.e. preferential flow, could not be ignored at all (Fig. 7). One possible mechanism might be that new micro- or meso-pores between the root surface and the surrounding soil particles increasingly form during root growth, leading to active water flow events (Jin et al., 2017; Franklin et al., 2021). Those pores are created around the root surface by tension (Ghestem et al.,

2011). It was found that the tension can reduce the root diameter and create a void around the root surface, which allows for preferential flow development (Chigira, 2001; Ghestem et al., 2011). Another mechanism is that root systems can form preferential root channels, which can establish connectivity between existing channels and make them more hydrologically active (Benard et al., 2021; Zhou et al., 2023). Those root channels created from the decomposition of root systems directly enhance soil pore formation, which also lead to active water flow events (Zhang et al., 2021). In addition, the direct effect of root properties on IWFC in the WFZ pattern is highly promoted, which is caused by the positive direct effects of root properties on water flow dynamics both in the PFP and SBZ patterns (Fig. 7). In summary, root properties can positively affect the entire water flow events, i.e., fast- and slow convective process, during root growth, which is supported by the results of Keleş (2019) and Jiang et al. (2021).

4.2. Soil properties and water flow

Consistent with previous studies of Celik (2005) and Jiang et al. (2021), the results showed that SOC was significantly positively correlated with TSP, SNCP, TN, TP, AN, and AP ($P < 0.01$), but negatively correlated with SBD ($P < 0.01$) (Fig. 6b). This is because SOC can increase the stability of soil aggregate structure (Adhikari and Bhattacharyya, 2015; Védère et al., 2022). In addition, SOC can reduce soil compaction and improve soil hydraulic properties (Bottinelli et al., 2010; Blanco-Canqui and Benjamin, 2013). Our results showed that soil chemical properties positively affect soil physical properties (Fig. 7), which is also attributed from the improvement of soil aggregate stability. The results are consistent with Hati et al. (2015) and Adhikari and Bhattacharyya (2015). In this study, TSP, silt, and ISWC were selected as the most important soil physical parameters controlling water flow connectivity (Fig. 4), due to soil porosity's key indicator of water flow path size and continuity (Wang et al., 2023), silt's reshaping the formation of soil aggregates (Wang et al., 2023), and soil water content's affecting soil water potential (Draye et al., 2010; Zhu et al., 2019). All of which could primarily affect soil infiltration and water flow dynamics.

In agreement with the results of Zhang et al. (2021), soil physical properties had the largest positive effects on water flow connectivity in the PFP pattern, while a negative effect in the SBZ pattern (Fig. 7). The present results indicated that TSP was treated as the selective representative indicator (Table 2, Fig. 4), which could reflect the abundance of soil macropores. Essentially, it should be evident that abundant soil macropores will promote the occurrence of preferential flow (Weiler and Naef, 2003), which can establish connectivity between those existing preferential flow paths (Wu et al., 2022). Our results found that TSP and IWFC in the PFP pattern are the largest in the bamboo forest, while they are the lowest in the oak forest, which dramatically supports our hypothesis (Table 2, Fig. 5). The present results also indicated that silt was the selective representative indicator due to its dominant fraction of soil particles (accounting for 80–90 %) in this study area (Table 2). Silty particles as cementing agents can result in soil aggregates with a higher mean weight diameter (Basset et al., 2023). Air space macropores would increase with increasing aggregate diameter, promoting soil water repellency (Menon et al., 2020). Similarly, more macroporous structural domains are present in large aggregates (Menon et al., 2020), which results in the development of preferential flow. As a result, the soil is physically well structured due to the aggregates improvement and water flow can quickly enter the deeper soil layers (Tang et al., 2019; Chen et al., 2021). In final, initial soil water content as the third selected indicator of soil physical properties could also positively affect the occurrence of preferential flow in our study (Fig. 6a). However, our results are not consistent with Kobayashi and Shimizu (2007) and Williams et al. (2020). The present results are supported by the correlation analysis (Fig. 6). In the present findings, the lowest initial soil water content was found in the pine forest, resulting in the lowest level of water flow connectivity in the PFP pattern, which also supports our

hypothesis. Previous studies concluded that a higher initial soil water content can increase the depth of soil macropore flow penetration (Wu et al., 2022). This is because a higher initial soil water content can lead to a decreased lateral- and an increased longitudinal infiltration capacity (Weiler and Naef, 2003), and also a lower sorption capacity of the soil matrix (Sanders et al., 2012). These can reduce the lateral flow into the surrounding soil matrix (Zambon et al., 2021; Wu et al., 2022).

4.3. Comparison of the impact of root and soil properties driven by SOC

4.3.1. Direct impacts

In agreement with the results of Zhang et al. (2021), our study showed that the total and direct effects of soil physical properties on the water flow connectivity in the PFP pattern is higher than that of root properties, while it is opposite in the SBZ pattern (Fig. 7). In the present study, parameters of root systems besides root tortuosity were positively correlated with soil porosity (Fig. 6b). Root systems can increase the formation of preferential flow paths by the development of root channels, which is supported by the results of Hoang et al. (2016). Another possible mechanism might be that the accumulation of soil nutrients can promote the stability of soil aggregates (Jiang et al., 2019; Védère et al., 2022). The present results indicated that soil nutrients content was positively with soil porosity (Fig. 6b), which supported our hypothesis. The combination of root systems and soil nutrients leads to the improvement of soil macropores structure (Franklin et al., 2021; Védère et al., 2022), and consequently making the existing preferential flow paths hydrologically active. However, there are limited number of channels formed by decayed root systems so that active water flow events through those channels are not dramatically obvious (Hoang et al., 2016). A large number of living root systems mainly absorb water and nutrients to maintain the vegetation productivity and thereby inhibit the smooth water flow from soil surface into the deep soil layers (Ghestem et al., 2011). Therefore, the effects of soil physical properties on active water flow events were significantly stronger than root properties in the PFP pattern (Fig. 7a, b).

Consistent with the results of Hao et al. (2021), root growth in the SBZ pattern can lead to the connectivity or tortuosity of soil pores. This is because root systems not only grow into the soil macropores along with the pore walls, but also pass through the compacted soil matrix (Ghestem et al., 2011), which increases the chance of water flow contacting with the surrounding soil matrix (Jin et al., 2017; Carminati et al., 2017). Our results showed that root properties have a higher effect on the water flow connectivity in the SBZ pattern (Fig. 7), which promotes the stream buffer zone effect. The results are supported by Jiang et al. (2021). The stream buffer zone is more compacted compared with the locations with lots of preferential flow paths for its small capillary pores (Zhang et al., 2017; Franklin et al., 2021), which results in the remarkably weakened effects of soil properties in those compacted locations on active water flow events. Therefore, root systems have the prominent effects on the hydrological responses of the stream buffer zones (Fig. 7c, d).

4.3.2. Indirect impacts

In the present study, it was found that SOC can positively affect the effectiveness of water flow connectivity in the three patterns (PFP, SBZ, and WFZ) (Fig. 7). However, the effects of SOC through root- and soil properties were different (Fig. 7). One possible mechanism might be that SOC can lower soil surface sealing and thereby increase soil porosity (Rawls et al., 1983), enhancing water flow effectiveness (Védère et al., 2022). Another possible mechanism is that high content of SOC can increase soil water availability and regulate the soil structure (Murphy, 2015; Libohova et al., 2018). However, the total effects of SOC on water flow connectivity in the SBZ pattern tend to be ignored to some degree and remarkably be lower than that in the PFP and WFZ patterns (Fig. 7c, d). This is because stream buffer zones with a higher sorption capacity and a larger amount of soil micropores can extremely inhibit the water

flow compared with the promoting water flow events of those pores (Nunan et al., 2003; Franklin et al., 2021; Jiang et al., 2023). With respect to the water flow in the PFP pattern, the soil locations with a higher content of SOC create a hydrophobic soil environment, increasing the chance of active water flow through preferential flow paths (Bogner et al., 2010; Nimmo, 2012). Therefore, although the content of SOC in the upper soil layers is high, the hydrophobicity it brings makes it difficult for water flow to laterally infiltrate into the surrounding soil matrix (Bogner et al., 2010; Nimmo, 2012). The content of SOC decreases with the increase in soil depth, which leads to the decrease of hydrophobicity and also the decline in water flow effectiveness (Table 2).

4.4. Implications and future considerations

The soil water flow is important for water budgeting and sustainable forestry management (Keleş, 2019). The results concluded that SOC-induced root and soil properties complexity has quantitatively crucial effects on soil water dynamics. In this study, soil physical properties have a dramatically positive impact on the connectivity of preferential flow paths than root properties, while root properties have the greatest effects on the connectivity of slow convection transport (Fig. 6). In agreement with previous studies, soil water dynamics both in the preferential flow paths and the soil matrix can highly influence the sustainable forestry management. This is because, on the one hand, preferential flow paths can regulate the spatial distribution of soil chemical elements and affect the soil environment. On the other hand, the soil matrix can influence the soil hydraulic redistribution (i.e. transfer of soil water from moist to dry layers by root systems), which can prevent the occur of water stress, inhibit the loss of root xylem conductivity, and prolong chemical elements uptake of root systems during drought periods (Neumann and Cardon, 2012). These joint regulations of preferential flow paths and the soil matrix can control soil hydrology and are important for understanding soil water dynamics in forest ecosystems. In addition, the results confirmed the hypothesis reported by Wu et al. (2022) that root systems may promote preferential flow by mediating soil physical properties (i.e., soil macropores development). In fact, the three forest stands in this study were conducted under long-term enclosure protection conditions, which results in complex vegetation structures, high vegetation density, and large root biomass. Consequently, the number of connected macropores dominated by well-established root systems are increasingly improved, leading to the high permeability in the soil profiles. Some studies have shown that this permeable water may be redistributed and stored in the unsaturated zones of the soil matrix, which plays a crucial role in soil water supplement in our study area.

Previous studies showed that SOC plays an important role in forest ecosystem production (Georgiou et al., 2022). Our results concluded that SOC has a positive impact on the soil water dynamics, such as water flow connectivity (Fig. 7), which can provide a good guidance for the available selection of plantation forests. Forestry managers can increase the ecological niche to increase the carbon source after the development of plantation forests complexity and the litter input, which can promote water flow connectivity at the root-soil interface (Cai et al., 2024). However, the distribution of various forms of SOC (i.e., dissolved organic carbon (DOC), particulate organic carbon (POC), minor-associated organic carbon (MOC), microbial organic carbon (MBC), and so on) with soil depth is different. These various forms of SOC are spatially autocorrelated and have different stability characteristics (Keskin et al., 2019). Therefore, determining which fractions of SOC have a greater impact on soil water dynamics will increasingly be of interest.

5. Conclusions

In this study, the authors tried to answer the hypothesis that SOC-

induced root and soil properties complexity contributes differently to the water flow connectivity. This study yielded several key findings:

- (1) *IWFC* in the PFP and WFZ patterns decreased with increasing soil depth, while *IWFC* in the SBZ pattern increased at first and then decreased.
- (2) The changes of *IWFC* stems from two direct mechanisms (changes in soil physical- and root properties) and indirect mechanisms (changes in the content of SOC).
- (3) Soil physical properties exerted a positively stronger impact on the active water flow events compared with root properties in the PFP pattern, while root properties showed the largest positive effects in the SBZ pattern. Both of them could equally affect the water flow connectivity in the WFZ pattern.
- (4) The influences of SOC by driving the changes of soil properties on gravity-driven convective flow process were dramatically stronger than root properties in the PFP and WFZ patterns, while SOC could primarily drive the changes of root properties and thereby affect capillary-driven convective flow process in the SBZ pattern.

The results provide new perspectives for distinguishing the role of root and soil properties in affecting the water flow. This study can provide tools for better improving hydrological functions of forest ecosystems.

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CRediT authorship contribution statement

Zhiying Tang: Writing – original draft, Visualization, Software, Resources, Investigation, Data curation. **Wenqi Zhang:** Validation, Software, Resources, Data curation. **Jinhong Chen:** Validation, Resources, Data curation. **Lu Wang:** Supervision, Resources, Data curation. **Yinghu Zhang:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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